BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – IST

Paper II (Objective Type) — Ist - A - Exam 2023

Time Allowed: 20 Minutes | **Maximum Marks:** 17

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. Mark the Correct Statement :

(A) All Lanthanides are present in the same group

(B) All Halogens are present in the same period

(C) All the Alkali Metals are present in the same group

(D) All the noble gases are present in the same period

Q2. Chile Saltpetre has the Chemical Formula :

(A) NaNO3

(B) KNO3

(C) Na2B4O7

(D) Na2CO3.H2O

Q3. Tincal is a Mineral of :

(A) Al

(B) B

(C) Si

(D) C

Q4. Oxidation of NO in air produces :

(A) N2O

(B) N2O3

(C) N2O4

(D) N2O5

Q5. Which Halogen occurs naturally in a positive Oxidation state :

(A) Fluorine
(B) Chlorine
(C) Bromine
(D) Iodine

Q6. The strength of binding energy of transition elements depends upon :

(A) Number of Electron pairs
(B) Number of Unpaired Electrons
(C) Number of Neutrons
(D) Number of Protons

Q7. Linear Shape is associated with which set of Hybrid Orbitals :

(A) sp
(B) sp2
(C) sp3
(D) dsp2

Q8. Preparation of Vegetable Ghee involves :

(A) Halogenation
(B) Hydrogenation
(C) Hydroxylation
(D) Dehydrogenation

Q9. The Electrophile in Aromatic Sulphonation is :

(A) H2SO4
(B) HSO4-
(C) SO3
(D) SO3+

Q10. Elimination Bimolecular reactions involve :

(A) First Order Kinetics
(B) Second Order Kinetics
(C) Third Order Kinetics
(D) Zero Order Kinetics

Q11. Methyl Alcohol is not used :

(A) As a Solvent
(B) As an Anti-freezing Agent
(C) As a substitute for Petrol
(D) For Denaturing of Ethyl Alcohol

Q12. The other name of Phenol is :

(A) Carbonic Acid
(B) Carbolic Acid
(C) Carboxylic Acid
(D) Adipic Acid

Q13. Which One will not give Silver Mirror Test :

(A) Formaldehyde
(B) Acetaldehyde
(C) Acetone
(D) Propionaldehyde

Q14. The number of Peptide Bonds in a Dipeptide is :

(A) 1
(B) 2
(C) 3
(D) 4

Q15. Select the Monomers of Nylon - 6, 6 :

(A) Adipic Acid and Ethylene Glycol
(B) Acetic Acid and Hexamethylene Diamine
(C) Adipic Acid and Hexamethylene Diamine
(D) Acetic Acid and Ethylene Glycol

Q16. Ammonia contains how much percentage of Nitrogen :

(A) 50 %
(B) 82 %
(C) 46 %
(D) 100 %

Q17. Ozone Layer is present in which region of Atmosphere :

(A) Troposphere
(B) Stratosphere
(C) Thermosphere
(D) Ionosphere

Chemistry (Subjective) — Ist - A - Exam 2023

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: It is compulsory to attempt any (8 - 8) Parts each from Q.No. 2, Q.No.3 and attempt any (6) Parts from Q.No.4. Attempt any (3) Questions from Part - II. Write same Question No. and its Part No. as given in the Question Paper — Part - I — 22 x 2 = 44 — Make Diagram where necessary.

Q2.

Q.No.2

(i) Write any two points in which Boron is different from its family members.

(ii) Why is increase in Atomic Size not regular in case of Group III-A elements?

(iii) What are Silicones? Give their any two properties.

(iv) What are Fused Membered Polycyclic Compounds? Draw structure of two compounds.

(v) What do you mean by Side Chain Oxidation of Toluene?

(vi) How did Kekule support his theory about structure of Benzene? Give any two points.

(vii) What are Thermoplastic Polymers? Give one example.

(viii) What are Polysaccharides? Give two examples.

(ix) What is Saponification Number? Give one example.

(x) Which human activities are responsible for the release of Hydrocarbons in air?

(xi) How do the Detergents cause the Water Pollution?

(xii) How is Acid Rain harmful for Aquatic Animals?

Q3.

Q.No.3

(i) What is meant by Hybridization?

(ii) Write down the structures of Isomers having Molecular Formula C4H10.

(iii) How is Ethyne Polymerized to Neoprene?

(iv) How can you differentiate between Saturated and Unsaturated Hydrocarbons?

(v) How is Ethane produced from Ethyl Magnesium Bromide?

(vi) Why is SO3 dissolved in H2SO4 and not in water?

(vii) Write down the reaction of Cu with: (a) dil. HNO3 (b) Conc. HNO3

(viii) How is H3PO3 produced from PCl5?

(ix) Write any two methods for the preparation of Alkyl Halides from Alcohol.

(x) Write down the reaction of Ammonia with Ethyl Bromide.

(xi) What are Essential Qualities for a Good Fertilizer? (Any Four)

(xii) Write down any four Non-Woody Raw Materials for Paper Manufacturing.

Q4.

Q.No.4

(i) Define Paramagnetism and Diamagnetism.

(ii) What will happen when KMnO4 reacts with: (a) Oxalic Acid (b) KOH

(iii) What are the Substitutional Alloys?

(iv) How can we distinguish between Ethanol and Methanol?

(v) What will happen when Diethyl Ether react with HI?

(vi) Write structural formula of Glycol and Glycerol.

(vii) What product would you obtain when Acetaldehyde reacts with 2,4-DNPH?

(viii) Write the Silver Mirror Test.

(ix) Write down reaction of Acetic Acid with: (a) Thionyl Chloride (b) PCl5

> Note: Part - II — 3 x 8 = 24

Q5. (Marks 8) (a) Explain the position of Hydrogen in I-A and VII-A Groups. (4) (b) What is the role of Gypsum in Agriculture? (4)

Q6. (Marks 8) (a) Discuss the Beckmann's Method for the preparation of Bleaching Powder. (4) (b) Write down the Bleaching Step involved in the manufacturing of Paper Process. (4)

Q7. (Marks 8) (a) Why there is no free rotation around a double bond and a free rotation around a single bond? Discuss Cis-trans Isomerism. (4) (b) What happens when: (a) Benzene is heated with Conc. H2SO4 at 250 deg C. (b) Chlorine is passed through Benzene in Sunlight. (c) A mixture of Benzene vapours and air are passed over heated Vanadium Pentoxide. (d) Benzene is burnt in free supply of Air. (4)

Q8. (Marks 8) (a) How will you Synthesize the following compounds starting from Ethyne: (a) Benzene (b) Ethane (c) Oxalic Acid (d) Methyl Cyanide (4) (b) Discuss SN1 reactions in detail. (4)

Q9. (Marks 8) (a) How Carbonyl Compounds are reduced by NaBH4? Give the Mechanism also. (4) (b) How will you prepare Carboxylic Acids from: (i) Hydrolysis of Esters (ii) Oxidative Cleavage of Alkenes (iii) Alkane Nitrile (iv) Alcohols (4)

GROUP – 2ND

Paper II (Objective Type) — Ist - A - Exam 2023

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8482

> Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. Oxides of Beryllium are :

(A) Acidic
(B) Basic
(C) Amphoteric
(D) Neutral

Q2. The number of Elements classified by Newlands in the Periodic Table are :

(A) 38
(B) 62
(C) 92
(D) 85

Q3. Which of the following elements is not present abundantly in Earth's Crust :

(A) Silicon
(B) Aluminium
(C) Sodium
(D) Oxygen

Q4. Oxidation of NO in air produces :

(A) N2O
(B) N2O3
(C) N2O4
(D) N2O5

Q5. Hydrogen Bond is the strongest between the Molecules of :

(A) HF
(B) HCl
(C) HBr
(D) HI

Q6. f-block elements are also called :

(A) Non Typical Transition Elements
(B) Outer Transition Elements
(C) Normal Transition Elements
(D) Inner Transition Elements

Q7. The Linear Shape is associated with which set of Hybrid Orbitals :

(A) sp
(B) sp2
(C) sp3
(D) dsp2

Q8. Preparation of Vegetable Ghee involves :

(A) Halogenation
(B) Hydrogenation
(C) Hydroxylation
(D) Dehydrogenation

Q9. The conversion of n-hexane into Benzene by heating in the presence of Pt is called :

(A) Isomerization
(B) Aromatization
(C) Dealkylation
(D) Rearrangement

Q10. Grignard Reagent is reactive due to :

(A) The presence of Halogen Atom
(B) The presence of Mg atom
(C) The polarity of C-Mg bond
(D) None of these

Q11. Which Enzyme is not involved in fermentation of Starch :

(A) Diastase
(B) Zymase
(C) Urease
(D) Maltose

Q12. The Carbon Atom of a Formaldehyde is :

(A) sp-Hybridized
(B) sp2-Hybridized
(C) sp3-Hybridized
(D) dsp2-Hybridized

Q13. Acetic Acid contains :

(A) A Hydroxyl Group
(B) A Carboxyl Group
(C) A Hydroxyl and Carboxyl Group
(D) A Carboxyl and an Aldehydic Group

Q14. Which of these Polymers is a Condensation Polymer :

(A) nylon-6,6
(B) Polystyrene
(C) Polyethene
(D) Polyvinylchloride

Q15. Which Woody Raw Material is used for the manufacture of Paper Pulp :

(A) Eucalyptus
(B) Bagasse
(C) Wheat Straw
(D) Bamboo

Q16. Which One is most Toxic :

(A) CO2
(B) CO
(C) O2
(D) CH4

Q17. Ethanol can be converted into Ethanoic Acid by

(A) Hydrogenation
(B) Hydration
(C) Oxidation
(D) Fermentation

Chemistry (Subjective) — Ist - A - Exam 2023

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: It is compulsory to attempt any (8 - 8) Parts each from Q.No. 2, Q.No.3 and attempt any (6) Parts from Q.No.4. Attempt any (3) Questions from Part - II. Write same Question No. and its Part No. as given in the Question Paper — Part - I — 22 x 2 = 44 — Make Diagram where necessary.

Q2.

Q.No.2

(i) How will you convert Boric Acid into Borax and Vice Versa?

(ii) Why is CO2 a gas at room temperature while SiO2 is a Solid?

(iii) What are Semiconductors? Give two examples.

(iv) Write two Electrophilic Substitution Reactions of Benzene.

(v) Prepare Benzene from Acetylene.

(vi) Draw the resonance structures of Benzene.

(vii) Differentiate between Thermoplastic and Thermosetting Polymers.

(viii) Write structures of Monomers of Epoxy Resins.

(ix) How are Proteins classified based on their functions?

(x) Describe Hazards of Ozone as a Pollutant.

(xi) What do you mean by Biochemical Oxygen Demand (BOD)?

(xii) Write the two harmful effects of Chlorination of Water.

Q3.

Q.No.3

(i) Write down names and formulas of any two Ores of Sulphur.

(ii) How is H2SO4 used to Dehydrate the Oxalic Acid and Glucose?

(iii) How does Aqua Regia dissolve gold in it?

(iv) What is Reforming? Give its chemical equation.

(v) What are Alicyclic Compounds? Draw structure of two compounds.

(vi) What is Markownikov's Rule? Give one example.

(vii) Why are Alkenes more reactive than Alkanes?

(viii) How will you prepare Chloroprene from Ethyne?

(ix) Give two methods for preparation of Alkylhalides from Alcohol.

(x) What are SN2 Reactions? Give one example.

(xi) How is Diammonium Phosphate prepared?

(xii) Write down names of any four Non-Woody Raw Materials for Paper.

Q4.

Q.No.4

(i) Under what conditions does Aluminium Corrode?

(ii) What is meant by Chelates?

(iii) Why are Transition Metals tough?

(iv) How can we distinguish Ethanol and Methanol?

(v) What is meant by Denaturing of Alcohol?

(vi) How is Phenyl Acetate produced from Phenol?

(vii) Write down any four uses of Acetaldehyde.

(viii) How can you distinguish Carbonyl Compounds from Non-Carbonyl Compounds?

(ix) What is Zwitter Ion?

> Note: Part - II — 3 x 8 = 24

Q5. (Marks 8) (a) What are the Halides? Write its types. Describe one of them in detail. (4) (b) What will happen when: (i) Lime is heated with coke (ii) Calcium

Nitrate is heated (iii) Gypsum is heated (iv) Beryllium Oxide is treated with NaOH (4)

Q6. (Marks 8) (a) Explain Peculiar Behaviour of Fluorine. (4) (b) What is Role of Digestion Step in the manufacture of Paper Pulp? (4)

Q7. (Marks 8) (a) Define sp3 Hybridization. Explain formation of Ethane Molecule according to this approach. (4) (b) Describe the structure of Benzene on the basis of Atomic Orbital Treatment. (4)

Q8. (Marks 8) (a) Write down the mechanism of Kolbe's Electrolytic Method for the preparation of Ethane. (4) (b) Discuss SN1 reactions in detail. (4)

Q9. (Marks 8) (a) What is Haloform Reaction? How will you prepare Iodoform from: (a) Acetaldehyde (b) Acetone (c) Ethanol (4) (b) Write Mechanism for the formation of Ethyl Acetate from Ethyl Alcohol and Acetic Acid in the presence of H2SO4. (4)